

PRANVEER SINGH INSTITUTE OF TECHNOLOGY KANPUR

Odd Semester

Session 2025-26

CT-I

B. Tech. - 5th Semester**Data Analytics (BCS052/ BCS052H)**

CO	Course Outcome
CO1	Define [L1: Remember] the fundamental concepts of data analytics, such data preprocessing, and basic statistical measures.
CO2	Describe [L2: Understand] the data analytics processes are used to transform raw data into actionable insights.
CO3	Apply [L3: Apply] basic data analysis & Visualization using programming languages (e.g., Python or R) and libraries such as Pandas, NumPy, and Matplotlib.
CO4	Analyzing [L4: Analyze] Compare, and evaluate different data analytics models in terms of accuracy, efficiency, and suitability for solving specific problems.

M. M. 20

Time: 1.5 Hrs.

Section A

(1X5 = 5 Marks)

Q1. Attempt all questions:

- Specify the big data.
- Recognize the type of data analytics, one will use for recommendation system
- List the common open source tools for the model building phase.
- Define Regression analysis.
- Explain multivariate analysis.

CO2
CO2
CO1
CO1
CO2

Section B

(2.5X4 = 10 Marks)

Q2. Attempt all questions:

- Describe reporting and analysis.
- Describe data analysis and data analytics.
- Explain the nature of data in data analytics.
- Explain the sources of data.
- Illustrate the characteristics of Big Data.
- Illustrate the common needs of Data analytics
- Differentiate between structured data, unstructured data and semi-structured data.
- Differentiate between descriptive data analytics and diagnostic data analytics.

CO2
CO2
CO2
CO3
CO3
CO2
CO2

Section C

(5X1 = 5 Marks)

Q3. i) Analyze various phases of Data Analytics Life Cycle.

CO4

ii) The sales of a company (in million dollars) for each year are shown in the table below:

CO4

x (year): 2005 2006 2007 2008 2009

y (sales): 12 19 29 37 45

- Use the regression line as a model to estimate the sales of the company in 2012.
- Investigate the coefficient of determination (R^2).

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CT-I

B. Tech.- 5th Semester

Web Technology (BCS502/BCS502H)

Course Outcome	
CO	
CO1	Able to define [L1: Remember] Web Development Strategies, Internet Tools and the protocols that govern the web, Internet Services, basic HTML tags and CSS.
CO2	Able to describe [L2: Understand] web development strategies, architecture of the client-server computing, concepts of JavaScript, XML, Java Networking APIs, Enterprise Java Beans, AJAX, Servlets and JSP.
CO3	Able to demonstrate [L3: Apply] the concepts of HTML, CSS, JavaScript, Node.js, Express Framework, Servlet, JSP, Java Networking concepts such as Internet addressing & TCP/IP Sockets, and database operation using MongoDB.
CO4	Able to examine [L4: Analyze] the different kinds of architecture of web, XML Parsers, different types of CSS, JavaScript and various advance JavaScript concepts for server-side programming, RESTful APIs and Beans.
CO5	Able to compare [L5: Evaluate] dynamic and client & server-side interactive web applications.

Time: 1.5 Hrs.

M. M. 20

Section A

(1X5 = 5 Marks)

Q1. Attempt all questions:

- State the history of web with a brief about each era.
- List the different attributes and tags of Frames in HTML citing an example for the same.
- State any two web-based protocols used in communication.
- Describe any two internet services used to exchange information.
- Explain different ways of object creation in JavaScript.

CO1
CO1
CO1
CO2
CO2

Section B

(2.5X4 = 10 Marks)

Q2. Attempt all questions:

- Explain any two web development strategies to be followed by a web developer.
- OR
- Differentiate between DOM and SAX processors of XML.
- Explain about functions in JavaScript, stating different ways of creating a function.
- OR
- Explain about the client-server TCP/IP sockets by citing an example showing communication between them.
- Express AJAX in detail and debate on the advantages and disadvantages of it.
- OR
- Explain about different pop-up boxes available in JavaScript citing example of each with syntax.

CO2
CO2
CO2
CO2
CO2